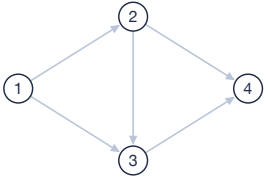
a A road map



five one-way roads;
no road leaves town 4

b walks of 1 road

A

	1	2	3	4
1	0	1	1	0
2	0	0	1	1
3	0	0	0	1
4	0	0	0	0

1 to 4: 0

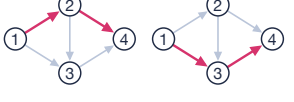
no road goes
straight from 1 to 4

c walks of 2 roads

A^2

	1	2	3	4
1	0	0	1	2
2	0	0	0	1
3	0	0	0	0
4	0	0	0	0

1 to 4: 2

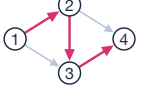


d walks of 3 roads

A^3

	1	2	3	4
1	0	0	0	1
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

1 to 4: 1



Entry (u, v) of A^ℓ counts the walks of ℓ roads from u to v. From 1 to 4: 0 + 2 + 1 = 3 walks in all, the (1, 4) entry of $I + A + A^2 + A^3$.